YU LIU

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Ph.D. applicant in Applied Mathematics / Scientific Computing — Fall 2027

RESEARCH INTERESTS

Numerical analysis; stochastic analysis; scientific computing. Monte Carlo methods, numerical approximation, and optimization.

EDUCATION

The Chinese University of Hong Kong, Shenzhen 2022–2024

M.Phil. in Mathematics | Cumulative GPA: 3.436/4.000.

Thesis: *Pricing and Trading Strategies of Chinese Convertible Bonds.* Advisor: Prof. Gongqiu Zhang.

Tongji University 2016–2020

B.Sc. in Mathematics and Applied Mathematics.

Major-course average: 85.09/100; overall equivalent average: 82.11/100 (official transcript).

Thesis: *Image Restoration: a Wavelet Frame Based Approach and Beyond.* Advisor: Prof. Jae Kyu Choi.

Selected A-grade coursework: Mathematical Analysis I–III, Real Analysis, Theory of Probability, Numerical Analysis (3 credits), Computational Method.

RESEARCH EXPERIENCE

Monte Carlo Methods for Path-Dependent Convertible Bond Valuation 2023–2024

M.Phil. research, CUHK-Shenzhen

- Modeled geometric Brownian stock-price dynamics and path-dependent call, put, and conversion-price reset provisions using Monte Carlo simulation.
- Applied least-squares continuation-value approximation within backward dynamic programming; studied the full-column-rank condition for uniqueness of regression coefficients.
- Investigated the effects of model assumptions, regression specifications, and exercise rules on numerical valuation. The full thesis also includes multi-factor strategy research.

Wavelet Frames and Numerical Optimization

Undergraduate thesis, Tongji University

- Studied analysis-based, synthesis-based, and balanced wavelet-frame reconstruction models and their connections to sparse regularization and soft thresholding.
- Worked through ADMM / Split Bregman formulations for the analysis-based model, FFT-based deblurring subproblems, and assumptions behind existing convergence results.
- Compared B-spline and Haar frame methods with total variation for deblurring and inpainting, evaluating PSNR, HFEN, SSIM, iteration counts, and runtime.

PREPRINT

Liu, Y. *Valuation Model of Chinese Convertible Bonds Based on Monte Carlo Simulation.* arXiv:2409.06496 (2024; updated 2026). Preprint; not peer-reviewed.

TEACHING EXPERIENCE

Teaching Assistant, MAT1012 Honours Calculus II 2024

The Chinese University of Hong Kong, Shenzhen

- Prepared and taught tutorials and exam-review sessions covering proof-based and computational problems; graded assignments, quizzes, and examinations; supported individual problem-solving discussions.

PROFESSIONAL EXPERIENCE

Backend Developer | Mavericksmobile, Chengdu Jan–Apr 2026

Contributed to AI application backends, task-processing workflows, monitoring, and maintenance.

Backend Software Developer | BYD, Shenzhen Sep 2024–Jul 2025

Developed backend services, content-management features, and workflow automation.

PROGRAMMING AND LANGUAGES

C++, Python, MATLAB, Java. English: IELTS 7.0.